The empirical recurrence for OEIS A303687, recovered from its tail

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8 September 2026

Abstract

OEIS A303687 records "Empirical recurrence of order 93" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 161$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{93} = 128 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A303687 is "Number of nX5 0..1 arrays with every element unequal to 2, 3 or 4 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$0, 61, 27, 343, 349, 2809, 4619, 29475, \ \ldots$$

The entry states:

> Empirical recurrence of order 93 (see link above)

As of the "Last modified" line on the live entry (Apr 28 2018 09:32:39, revision 5) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 161$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 161$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 322$ exact terms from $n = 5$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{93} c_i\, a(n-i),$$

c_1	0	c_{25}	-306557503	c_{49}	-6619594617	c_{73}	-382402090
c_2	37	c_{26}	-363264431	c_{50}	-19017220855	c_{74}	-141624352
c_3	6	c_{27}	549209385	c_{51}	2090846187	c_{75}	157772157
c_4	-610	c_{28}	638113805	c_{52}	16499920987	c_{76}	88078560
c_5	-166	c_{29}	-903056841	c_{53}	452879832	c_{77}	-49002106
c_6	6044	c_{30}	-824394297	c_{54}	-12016400524	c_{78}	-39288206
c_7	2189	c_{31}	1396257280	c_{55}	-774136629	c_{79}	10600206
c_8	-40686	c_{32}	625336677	c_{56}	7445458891	c_{80}	13456710
c_9	-18701	c_{33}	-2102990543	c_{57}	-26474938	c_{81}	-1112918
c_{10}	198309	c_{34}	258059418	c_{58}	-3892865158	c_{82}	-3609664
c_{11}	118478	c_{35}	3198395155	c_{59}	847289906	c_{83}	-204668
c_{12}	-723651	c_{36}	-1797499825	c_{60}	1557897420	c_{84}	752260
c_{13}	-598931	c_{37}	-4948229412	c_{61}	-1308286106	c_{85}	139704
c_{14}	1991657	c_{38}	3251257032	c_{62}	-187202264	c_{86}	-117464
c_{15}	2509373	c_{39}	7520567667	c_{63}	1496122244	c_{87}	-48368
c_{16}	-3958795	c_{40}	-3147843572	c_{64}	-462955119	c_{88}	10032
c_{17}	-8863731	c_{41}	-10613333681	c_{65}	-1520630718	c_{89}	12320
c_{18}	4292477	c_{42}	88269046	c_{66}	553775108	c_{90}	864
c_{19}	26627318	c_{43}	13167678654	c_{67}	1388534673	c_{91}	-1856
c_{20}	5798419	c_{44}	5921958071	c_{68}	-311038786	c_{92}	-256
c_{21}	-68678205	c_{45}	-13666709930	c_{69}	-1098127032	c_{93}	128
c_{22}	-47666315	c_{46}	-12880412675	c_{70}	20551693		
c_{23}	154200878	c_{47}	11216611518	c_{71}	722684432		
c_{24}	156759694	c_{48}	17873003349	c_{72}	133180946		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A303687 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1 x^{93-1} - \cdots - c_{93}$. Its constant term is $-c_{93} = -128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 128.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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